Stochastic Control

Homework Project 1

Reinforcement Learning with Policy Gradient

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Problem 1

a) The state space is $\mathbb{X}$ is defined by the set

$$\mathbb{X} = \{(y, z) | \, y \in \{1, \dots, 5\}, \, z \in \{1, \dots, 10\}\}.$$

b) The action space can be defined as follows

$$\mathbb{U} = \{s, \nu, d\},$$

where, assuming (y, z) is the current state of the the robot,

- s stands for *same*, i.e. the robot stays on the same y and moves only on the z direction with $z \leq 9$,
- ν stands for *up*, i.e. the robot increments both its y and z coordinates with $z \leq 9$ and $y \leq 4$,
- d stands for *down*, i.e. the robot decrements its y coordinate and increments the z coordinate with $z \leq 9$ and $y \geq 2$.

c) Let

$$\mathcal{B} = \{-r_{\text{obst}}, -r_{\text{diag}}, r_{\text{end}}\},$$

where $r_{\text{end}} \gg r_{\text{obst}} > r_{\text{diag}}$. The reward set $\mathcal{R}$ can be defined as follows

$$\mathcal{R} = \left\{ \sum_{x \in \mathcal{A}} x \, | \, \mathcal{A} \subseteq \mathcal{B} \right\}.$$

Then, $\mathcal{R}$ contains the sums of the elements of each subset of $\mathcal{B}$. Note that $0 \in \mathcal{R}$, as the sum of the elements of the empty subset. In this setting award

- $-r_{\text{obst}}$ when the robot steps on an obstacle cell,
- $-r_{\text{diag}}$ when the robot makes a diagonal movement, i.e. the action u is ν or d,
- r_{end} when the robot reaches a desired end position.

If more than one of the above conditions apply for a movement of the robot, then award the sum of the corresponding values. If none apply, then award 0.

d)